

CSN6214 – Tutorial 4

CPU Scheduling

1. Why is it important for the scheduler to distinguish I/O-bound programs from CPU-bound programs?
2. What is the difference between preemptive and non-preemptive scheduling?
3. Explain the difference in the degree to which the following scheduling algorithms discriminate in favor of short processes.
 - a) FCFS
 - b) RR
4. Consider the following set of processes, with the length of the CPU-burst time given in milliseconds:

Process	Burst-time	Arrival time
P1	10	0
P2	2	1
P3	3	2
P4	1	3
P5	5	4

Total = 21

- a) Draw the Gantt charts illustrating the execution of these processes using FCFS, SJN and RR (quantum=2) scheduling.
- b) What is the turnaround time of each process for each of the scheduling algorithms?
- c) What is the waiting time of each process for each of the scheduling algorithms?
- d) Which of the schedules in Q3 results in the minimal average waiting time (over all processes)?

5. Consider the following processes with the length of a CPU burst time given in milliseconds. The processes arrived according to the arrival time. (Low numbers represent high priority)

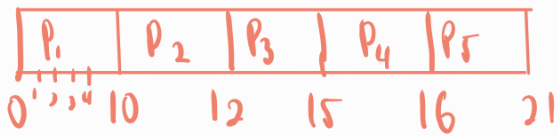
Process	Arrival Time	Priority	Burst Time
P0	0	2	8
P5	0	1	6
P1	4	5	15
P4	9	4	13
P2	7	3	9
P3	13	1	5

- a) Draw 3 (three) Gantt Charts illustrating the execution of these processors using scheduling.
 - i. Shortest Job Next
 - ii. Shortest Remaining Time
 - iii. Non - Preemptive priority scheduling
 - iv. Preemptive priority scheduling
 - v. Round Robin (quantum=3) scheduling
- b) Calculate the average waiting time for Preemptive priority and Round Robin Scheduling

Important must do

Test ques ex

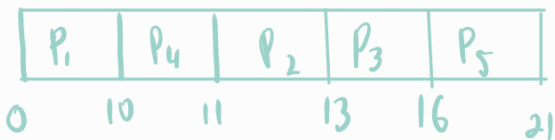
FCFS



- ~~0 - P₁(10)~~
- ~~1 - P₂(2)~~
- ~~2 - P₃(3)~~
- ~~3 - P₄(1)~~
- ~~4 - P₅(5)~~

SJN (Non-Prem)

Shortest Job Next

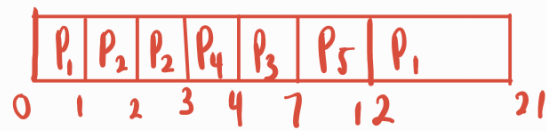


- ~~0 - P₁(10)~~
- ~~1 - P₂(2)~~
- 2 - P₃(3)
- 3 - P₄(1)
- 4 - P₅(5)

Turn around time = finish - arrival

Waiting time = turn around - burst time

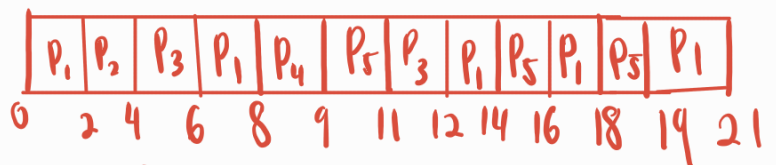
Pre-SJF (interrupted, every arr time)



- ~~0 - P₁(10)~~
- ~~1 - P₂(2)~~
- 2 - P₃(3)
- ~~3 - P₄(1)~~
- 4 - P₅(5)

RR

RR



- ~~0 - P₁(10)~~
- ~~1 - P₂(2)~~
- ~~2 - P₃(3)~~
- ~~3 - P₄(1)~~
- ~~4 - P₅(5)~~

Q5. total burst time = 56

SJN (short job first)

P ₅	P ₀	P ₃	P ₂	P ₄	P ₁
0	6	14	19	28	41

56

~~0 - P₀(8), P₅(6)~~

~~4 - P₁(15)~~

~~7 - P₂(9)~~

~~9 - P₄(13)~~

~~13 - P₃(5)~~

Pre-m - SJF (short first) (SRT)

P_5	P_5	P_0	P_0	P_0	P_0/P_3	P_2	P_4	P_1
0	4	6	7	9	13 14	19	28	41

56

~~0 - P₀(8), P₅(6)~~

~~4 - P₁(15)~~

~~7 - P₂(9)~~

~~9 - P₄(13)~~

~~13 - P₃(5)~~

Formula

Turnaround time = Finish time - Arrival time

Average TT = Sum of TT / Number of Processes

Waiting time = TT - CPU burst time

Average waiting time = Sum of WT / Number of Process.

Non-Prem Priority schd

P ₅	P ₀	P ₃	P ₂	P ₄	P ₁
0	6	14	19	28	41

56

~~0 - P₀(8)², P₅(6)¹~~
~~4 - P₁(15)⁵~~
~~7 - P₂(9)³~~
~~9 - P₄(13)⁴~~
~~13 - P₃(5)¹ ← high prio~~

Prem Priority schd

P ₅	P ₅	P ₀	P ₀	P ₃	P ₀	P ₂	P ₄	P ₁
0	4	6	7	9	13	18	19	28

41 56

~~0 - P₀(8)², P₅(6)¹~~
~~4 - P₁(15)⁵~~
~~7 - P₂(9)³~~
~~9 - P₄(13)⁴~~
~~13 - P₃(5)¹ ← high prio~~

Round Robin

Time Quantum = 3

RQ

P ₅	P ₀	P ₅	P ₁	P ₀	P ₂	P ₄	P ₁	P ₃	P ₀	P ₄	P ₁	P ₃	P ₂	P ₄	P ₁	P ₄	P ₁	P ₄
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P ₅	P ₀	P ₅	P ₁	P ₀	P ₂	P ₄	P ₁	P ₃	P ₀	P ₂	P ₄	P ₁	P ₃	P ₂	P ₄	P ₁	P ₄	P ₁	P ₄	
0	3	6	9	12	15	18	21	24	27	29	32	35	38	40	43	46	49	52	55	56

MS

~~0 - P₀(8)², P₅(6)¹~~
~~4 - P₁(15)⁵~~
~~7 - P₂(9)³~~
~~9 - P₄(13)⁴~~
~~13 - P₃(5)¹~~

b)

Preemptive Priority

P	FT	AT	TT	WT
P ₀	19	0	19	19 - 8 = 11
P ₁	56	4	52	52 - 15 = 37
P ₂	28	7	21	21 - 9 = 12
P ₃	18	13	5	5 - 5 = 0
P ₄	41	9	32	32 - 13 = 19
P ₅	6	0	6	6 - 6 = 0
				$\Sigma WT = 79$

$$\begin{aligned}
 \text{Avg. WT} &= \Sigma WT / P \\
 &= 79 / 6 \\
 &= 13.17 \text{ ms}
 \end{aligned}$$

RR

P	FT	AT	TT	WT
P ₀	29	0	29	29 - 8 = 21
P ₁	55	4	51	51 - 15 = 36
P ₂	43	7	36	36 - 9 = 27
P ₃	40	13	27	27 - 5 = 22
P ₄	56	9	47	47 - 13 = 34
P ₅	9	0	9	9 - 6 = 3

$$\Sigma WT = 143$$

$$\begin{aligned}
 \text{Avg. WT} &= 143 / 6 \\
 &= 23.83 \text{ ms}
 \end{aligned}$$